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CS302 Midterm Subjective Papers By Waqar Sidhu (File 1) Subjective

Question No : 21 of 26 Marks: 2 (Budgeted Time 4 Min)

How many possible outputs does a decoder of 3 inputs contain?

Answer (Please [click here](#) to Add Answer)

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Question No : 22 of 26 Marks: 2 (Budgeted Time 4 Min)

How can we use 3-to-8 decoder to implement SOP expressions?

Answer (Please [click here](#) to Add Answer)

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Question No : 23 of 26 Marks: 3 (Budgeted Time 6 Min)

For a two bit comparator circuit, specify all the inputs for which $A > B$ is set to 1.

Answer (Please [click here](#) to Add Answer)

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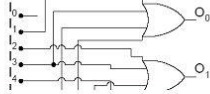




Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Identify the following circuit that is which combinational circuit is being represented by this diagram?



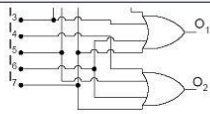
Answer (Please [click here](#) to Add Answer)

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Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)



Answer (Please [click here](#) to Add Answer)

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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Consider a logical circuit for 4-bit binary number. The circuit gives output 0 for adjacent 0s in a binary number. Draw the karnaugh map of such circuit and derive simplified POS expression.

Answer ([Please click here to Add Answer](#))

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Multiplexers can be used as a "Logic Function Generator". Draw only circuit diagram and function table based on three variables.

Answer ([Please click here to Add Answer](#))

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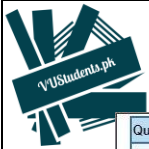
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(File 2)
Subjective



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Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

How many possible outputs does a decoder of 3 inputs contain?

Answer (Please [click here](#) to Add Answer)

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Decoder, the 3-to-8 Decoder has active-low outputs and three extra NOT gates connected at the three inputs to reduce the four unit load to a single unit load. The 3-to-8 Decoder has three enable inputs, one of the three enable inputs is active-high and the remaining two are active-low. All three enable inputs have to be activated for the Decoder to work

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Question No : 22 of 26

Marks: 2 (Budgeted Time 4 Min)

Once the Logic circuit design has been entered its operation is verified by using 'test vectors'. Write two major roles of test vectors.

Answer (Please [click here](#) to Add Answer)

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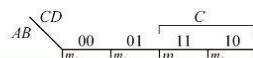
The testing verifies the design of the logic circuit before committing it to the PLD. If any flaws are detected during the testing process the design must be debugged and submitted for recompilation. Once the design has been finalized a documentation file is produced along with a fuse map file which is downloaded to the programmer which programs the PLD device inserted in the programmer socket.

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Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

Derive simplified Boolean equation from the following Karnuagh Map.



Answer (Please [click here](#) to Add Answer)

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Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

AB	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6

Answer ([Please click here](#) to Add Answer)

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Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

$$A = \left[\begin{array}{cc|cc|cc} & & & & & \\ & & & & & \\ 11 & m_{12} & m_{13} & m_{15} & m_{14} & \\ & & 1 & 1 & 1 & \\ 10 & m_8 & m_9 & m_{11} & m_{10} & \\ & & & 1 & & \end{array} \right] B$$

Answer ([Please click here to Add Answer](#))

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Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

The diagram shows a horizontal row of four rectangular boxes representing lattice sites. The boxes are labeled from left to right as m_8 , m_9 , m_{11} , and m_{10} . A double vertical line connects the top and bottom of the m_9 and m_{11} boxes. The m_{11} box also contains the number '1' in its center. Below the boxes, a horizontal bracket spans the width of the m_9 and m_{11} boxes and is labeled D . To the left of the first box, a vertical bracket labeled A indicates a height of 10.

Answer ([Please click here to Add Answer](#))

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Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Tri-state buffer control input can be connected in four different ways. Name any three of them.

Answer (Please [click here](#) to Add Answer)

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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Using Karnaugh map minimize the following POS expression:
 $(B + C + D)(A + B + C + D)(\bar{A} + B + C + D)(A + \bar{B} + C + D)(\bar{A} + \bar{B} + C + D)$

Answer (Please [click here](#) to Add Answer)

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Draw the function table of 8-to-3 Encoder.

Answer (Please [click here](#) to Add Answer)

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(File 3) Subjective

Question No : 21 of 26	Marks: 2 (Budgeted Time 4 Min)
How many possible outputs does a decoder of 3 inputs contain?	
Answer (Please click here to Add Answer)	
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Question No : 22 of 26	Marks: 2 (Budgeted Time 4 Min)
Write down four uses of multiplexers.	
Answer (Please click here to Add Answer)	
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Question No : 23 of 26	Marks: 3 (Budgeted Time 6 Min)
For a two bit comparator circuit, specify all the inputs for which the output $A = B$ is set to 1.	
Answer (Please click here to Add Answer)	
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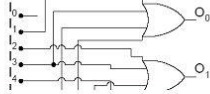




Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Identify the following circuit that is which combinational circuit is being represented by this diagram?



Answer (Please [click here](#) to Add Answer)

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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Map the following POS expression on a karnaugh map:

$$(\bar{A} + \bar{B} + C + D)(\bar{A} + B + \bar{C} + D)(A + B + \bar{C} + D)(\bar{A} + \bar{B} + \bar{C} + \bar{D})(A + B + \bar{C} + D)$$

Answer (Please [click here](#) to Add Answer)

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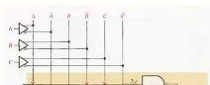
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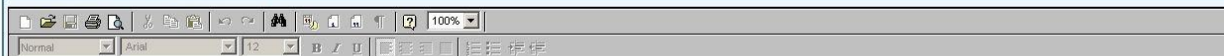
Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

A programmed array is shown in Figure below. The intact fusible link, are indicated by small red Xs. Absence of an X means that the fuse is open. Analyze the figure below and write the output expression.



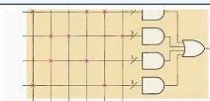
Answer (Please [click here](#) to Add Answer)



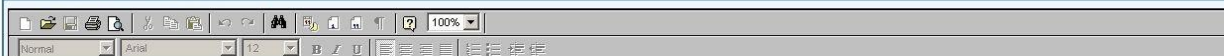
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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)



Answer ([Please click here to Add Answer](#))



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(File 4)
Subjective



Question No : 21 of 26

Marks: 2 (Budgeted Time 4 Min)

How many input and output bits do a Half-Adder contain?

Answer (Please [click here](#) to Add Answer)

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Question No : 22 of 26

Marks: 3 (Budgeted Time 4 Min)

Draw the Truth-Table of NAND based S-R Latch.

Answer (Please [click here](#) to Add Answer)

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Question No : 23 of 26

Marks: 3 (Budgeted Time 6 Min)

Draw the block diagram of 4-bit BCD Adder.

Answer (Please [click here](#) to Add Answer)

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Question No : 24 of 26

Marks: 3 (Budgeted Time 6 Min)

Tri-state buffer control input can be connected in four different ways. Name any three of them.

Answer (Please [click here](#) to Add Answer)

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Question No : 25 of 26

Marks: 5 (Budgeted Time 10 Min)

Simplify the following Boolean expressions, using Boolean Algebraic rules

- $(A+B)(A+B')$
- $(A+B)'(A'+B')'$

Answer (Please [click here](#) to Add Answer)

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Question No : 26 of 26

Marks: 5 (Budgeted Time 10 Min)

Multiplexers can be used as a "Logic Function Generator". Draw only circuit diagram and function table based on three variables.

Answer (Please [click here](#) to Add Answer)

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The END

